SAT Math Mastery · Session 8

Data Models, Angles and Triangles

Time	3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 45 minutes), and a short error log.
Topics	scatterplots and lines of best fit: predicted versus actual, slope and intercept in context, linear versus exponential models · parallel lines and transversals · angle sums and exterior angles · isosceles triangles and regular polygons · similar triangles and proportional segments · triangle inequality, midsegments and angle bisectors · right-triangle trigonometry and special right triangles · Law of Cosines
Name / Date	_____ _____

PART A · IN CLASS — 37 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type in Desmos, one expression per line; tap a grey dot to read an intersection, intercept or vertex. A letter such as a, c or k becomes a slider when you accept Desmos's offer to add one.

1. Medium · No calculator needed

A household water-conservation study recorded, for several summer days, the daily high temperature x (in degrees Celsius above 25 °C) and the household's water use y (in liters). A line of best fit for the data is given by $y = 12x+20$. On one recorded day, $x = 10$ and the household's actual water use was 155 liters. What is the positive difference between the actual water use on that day and the water use predicted by the line of best fit?

1. The line of best fit gives the predicted value. Substitute the given $x = 10$ into $y = 12x + 20$.

2. $y = 12(10) + 20 = 120 + 20 = 140$ liters is the predicted water use.

3. The actual water use was 155 liters. The positive difference is $155 - 140 = 15$.

Answer: 15

2. Medium · No calculator needed

A scatterplot shows the relationship between x, the distance in kilometers from a city's central waterfront, and y, the number of visitors counted per hour at a beach access point. The line of best fit for the data passes through the points $(2, 340)$ and $(10, 60)$. What is the slope of this line of best fit?

(A) 35 **(B) −35 ✓**

(C) −28 (D) 28

1. The slope is rise over run between the two points: $\dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{60 - 340}{10 - 2}$.

2. The numerator is -280 and the denominator is 8, so the slope is $\dfrac{-280}{8} = -35$.

3. The slope is negative because the visitor count falls as the distance grows. That is option (B); option 35 drops the sign and −28 or 28 come from a slip in the division.

Answer: (B)

3. Medium · Calculator

A line of best fit models y, the installed solar capacity (in megawatts) at a pilot site, in terms of x, the number of years since 2015: $y = 45.2x + 12.5$. Which of the following is closest to the capacity predicted by this model at $x = 8.4$?

(A) 380.5

(B) 388.7

(C) 392.2 ✓

(D) 401.0

1. Substitute $x = 8.4$ into $y = 45.2x + 12.5$. Multiply first, then add.
2. $45.2 \times 8.4 = 379.68$. Then $379.68 + 12.5 = 392.18$.
3. Rounded to one decimal place, $392.18 \approx 392.2$. That is option (C).

Answer: (C)

4. Medium · No calculator needed

A water-treatment plant's storage tank begins a shift completely full. The tank is drained at a constant rate throughout the shift to supply a neighborhood. What type of model best describes the volume of water remaining in the tank at any time during the shift?

(A) A linear model with a negative slope ✓

(B) An exponential growth model

(C) An exponential decay model

(D) A linear model with a positive slope

1. A constant rate means the same amount leaves the tank every unit of time. Example: a 1000-liter tank draining 85 liters per hour holds $1000, 915, 830, 745$ liters after $0, 1, 2, 3$ hours. The drops are 85 each time (constant).
2. Constant drops mean a linear model, and because the volume is going down the slope is negative.
3. Exponential decay removes the same percent each time, so the drops get smaller: 10% decay gives $1000, 900, 810, 729$ liters with drops 100, 90, 81 (not constant).
4. So the model is linear with a negative slope, option (A).

Answer: (A)

5. Medium · No calculator needed

A scatterplot shows exam-preparation hours x and practice-exam score y for 10 students, with a line of best fit shown. For the student with the highest practice-exam score, the actual score was 96, and the score predicted by the line of best fit was 89. Which of the following is closest to the difference between the actual score and the predicted score for this student?

(A) 4

(B) 1

(C) 10

(D) 7 ✓

1. The difference between the actual score and the predicted score is the actual value minus the predicted value (the vertical gap between the point and the line).
2. The actual score is 96 and the predicted score is 89.
3. $96 - 89 = 7$, so the point sits 7 points above the line. That is option (D).

Answer: (D)

6. Medium · Desmos

The table shows the age x, in years, and the efficiency loss y, in percent, for a sample of rooftop solar panels.

x	1	2	3	4	5
y	3.5	6.0	8.6	11.0	13.4

Which of the following equations could be the line of best fit for the data?

(A) $y = -2.5x + 1$

(B) $y = 2.5x - 1$

(C) $y = 2.5x + 1$ ✓

(D) $y = x + 2.5$

1. Open a table and enter $x_1 = 1, 2, 3, 4, 5$ and $y_1 = 3.5, 6, 8.6, 11, 13.4$.
2. Type `y_1~mx_1+b`: Desmos returns $m = 2.48$ and $b = 1.06$ (the line of best fit).
3. The slope is about $2.48 \approx 2.5$ and the intercept is about $1.06 \approx 1$, so the line is $y \approx 2.5x + 1$, option (C).
4. Check the others: option $y = x + 2.5$ has slope 1, option $y = -2.5x + 1$ slopes downward and option $y = 2.5x - 1$ has intercept -1, far from 1.06.

Answer: (C)

7. Hard · No calculator needed

In which of the following tables is the relationship between the values of x and their corresponding y-values nonlinear?

(A)

x	1	2	3	4
y	5	10	15	20

(B)

x	1	2	3	4
y	3	6	12	24

✓

(C)

x	1	2	3	4
y	5	9	13	17

(D)

x	1	2	3	4
y	2	4	6	8

1. A table is linear when y changes by the same amount for every 1-unit step in x. So check the differences between consecutive y-values.
2. The first table gives $4, 4, 4$, the second $2, 2, 2$ and the third $5, 5, 5$. Each is constant (4, 2 and 5), so all three are linear.
3. The fourth table gives $3, 6, 12$, which are not equal (each y-value is double the one before: a constant ratio, not a constant difference), so it is nonlinear. That is option (B).

Answer: (B)

8. HARD · NO CALCULATOR NEEDED

A line of best fit models the relationship between x, the number of years since 2018, and y, the average monthly electricity subsidy, in dollars, provided to households in an energy-efficiency pilot program: $y = -18.6x + 340$. Which of the following is the best interpretation of the slope of this line of best fit in this context?

(A) The estimated subsidy decreases by \$340 each year.

(B) The estimated initial subsidy was \$18.6.

(C) The estimated subsidy increases by \$18.6 each year.

(D) The estimated subsidy decreases by \$18.6 each year. ✓

1. In $y = mx + b$ the slope m is the change in y for each 1-unit increase in x. Here x is years since 2018 and y is the subsidy in dollars.
2. The slope is $m = -18.6$: from $x = 0$ to $x = 1$ the model goes from 340 to 321.4, a change of -18.6 dollars per year.
3. A negative change means the subsidy decreases by \$18.6 each year. The number 340 is the intercept: the estimated subsidy at $x = 0$ (the year 2018), not a rate. That is option (D).

Answer: (D)

9. HARD · DESMOS

Sam deposits \$800 into Account A, which earns 5% interest compounded annually. Sam also deposits \$1,500 into Account B, whose balance increases by a fixed \$40 each year. Based on these growth patterns, which of the following best describes the long-term behavior of the balances of Account A and Account B?

(A) The balance of Account A starts lower than the balance of Account B but eventually exceeds it. ✓

(B) The balance of Account A starts higher than the balance of Account B but eventually falls below it.

(C) The balance of Account A is always greater than the balance of Account B.

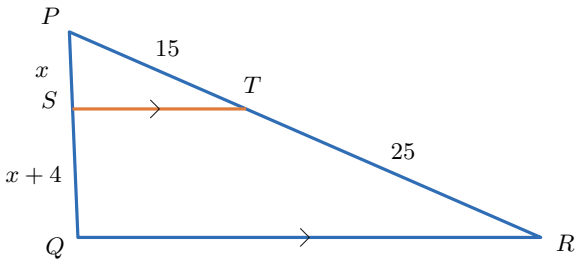
(D) The balance of Account A is always less than the balance of Account B.

1. Let x be the number of years. Account A is exponential: type `y=800(1.05)^x`. Account B is linear: type `y=1500+40x`.
2. Zoom out and tap the intersection with $x > 0$: (22.53, 2401.04). (The other intersection has $x = -33.6 < 0$, which is not a time after the deposits.)
3. At $x = 0$ the values are 800 and 1500, so A starts lower. After about 22.5 years the exponential curve is above the line and stays above it, because exponential growth eventually outpaces any linear growth. That is option (A).

Answer: (A)

10. HARD · NO CALCULATOR NEEDED

In triangle PQR, segment ST is drawn parallel to side QR, where S lies on PQ and T lies on PR. If $PS = x$, $SQ = x + 4$, $PT = 15$, and $TR = 25$, what is the value of x?



1. $ST \parallel QR$, so triangle PST is similar to triangle PQR and the sides are proportional: $\dfrac{PS}{SQ} = \dfrac{PT}{TR}$.

2. Substitute: $\dfrac{x}{x+4} = \dfrac{15}{25} = \dfrac{3}{5}$.

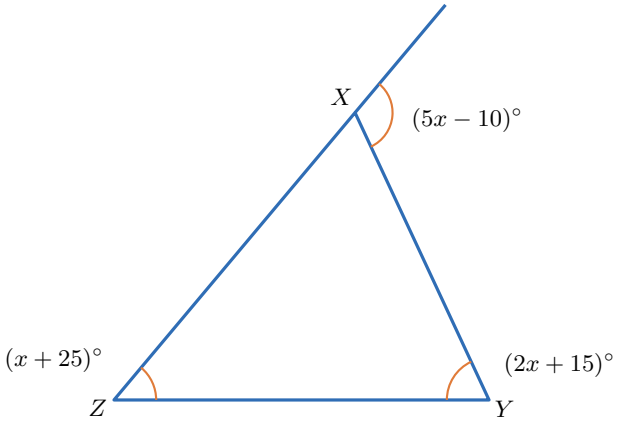
3. Cross-multiply: $5x = 3(x+4) = 3x + 12$, so $2x = 12$.

4. $x = 6$. Check: $PS = 6$ and $SQ = 10$, and $\dfrac{6}{10} = \dfrac{3}{5}$.

Answer: 6

11. HARD · NO CALCULATOR NEEDED

In triangle XYZ, the exterior angle at vertex X measures $(5x - 10)°$. This exterior angle equals the sum of the two non-adjacent (remote) interior angles, which measure $(2x + 15)°$ at vertex Y and $(x + 25)°$ at vertex Z. What is the measure of angle Y, in degrees?



1. The exterior angle equals the sum of the two remote interior angles: $5x - 10 = (2x + 15) + (x + 25)$.

2. Combine: $5x - 10 = 3x + 40$, so $2x = 50$ and $x = 25$.

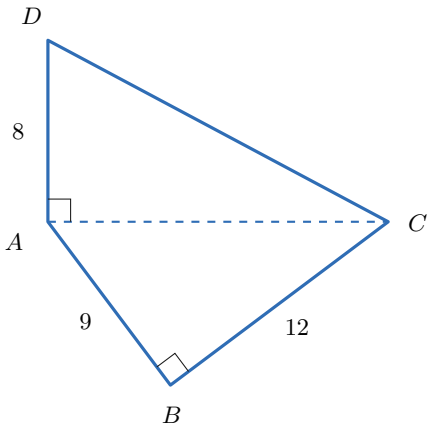
3. Angle $Y = 2(25) + 15 = 65°$.

4. Check: the exterior angle is $5(25) - 10 = 115°$ and $65 + 50 = 115$.

Answer: 65

12. HARD · NO CALCULATOR NEEDED

In the figure, quadrilateral $ABCD$ is formed by two right triangles, ABC and ACD, sharing diagonal AC. Triangle ABC has a right angle at B with $AB = 9$ and $BC = 12$. Triangle ACD has a right angle at A with $AD = 8$. What is the area of quadrilateral $ABCD$?



(A) 108

(B) 114 ✓

(C) 90

(D) 96

1. Triangle ABC has a right angle at B with legs 9 and 12, so $AC = \sqrt{9^2 + 12^2} = \sqrt{225} = 15$.

2. Its area is $\frac{1}{2}(9)(12) = 54$.

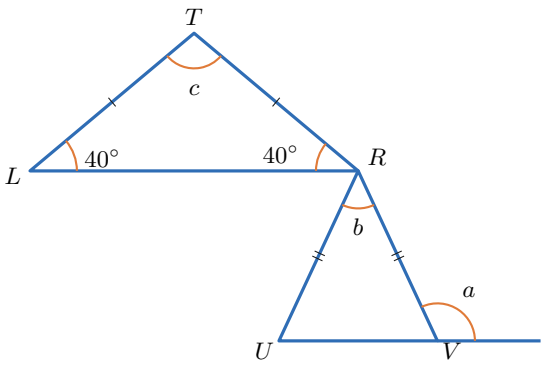
3. Triangle ACD has a right angle at A, so its legs are $AC = 15$ and $AD = 8$, and its area is $\frac{1}{2}(15)(8) = 60$.

4. Add the two areas: $54 + 60 = 114$. That is option (B).

Answer: (B)

13. HARD · NO CALCULATOR NEEDED

In the figure shown, two isosceles triangles share the vertex R. In the top triangle, the vertex angle is c and each of its two equal base angles measures $40°$. In the bottom triangle, the vertex angle b is at R and equals half of c. Angle a is formed by extending the base of the bottom triangle, so a is supplementary to one of the bottom triangle's two equal base angles. What is the value of a, in degrees?



1. Top triangle: its base angles are $40°$, so $c = 180 - 40 - 40 = 100°$.

2. The bottom vertex angle is half of c: $b = \dfrac{100}{2} = 50°$.

3. The bottom triangle is isosceles, so each base angle is $\dfrac{180 - 50}{2} = 65°$.

4. Angle a is supplementary to a base angle: $a = 180 - 65 = 115°$.

Answer: 115

14. HARD · NO CALCULATOR NEEDED

In triangle ABC, the exterior angle at vertex C measures $130°$. If the measure of angle A is $55°$, what is the measure of angle B?

1. The exterior angle at C equals the sum of the two remote interior angles: $130 = \angle A + \angle B$.

2. Substitute $\angle A = 55°$: $\angle B = 130 - 55 = 75°$.

3. Check with the angle sum: the interior angle at C is $180 - 130 = 50$, and $55 + 75 + 50 = 180$.

Answer: 75

15. HARD · NO CALCULATOR NEEDED

Lines l and m are parallel and are cut by transversal t. Which of the following pairs of measures could NOT represent the measures of two same-side interior angles formed by l, m, and t?

(A) 95° and 85°

(B) 112° and 68°

(C) 95° and 84° ✓

(D) 100° and 80°

1. Same-side interior angles between parallel lines are supplementary: they must add to 180°.

2. Add each pair: $100 + 80 = 180$, $95 + 85 = 180$, $112 + 68 = 180$ and $95 + 84 = 179$.

3. Only one sum differs from 180: $95 + 84 = 179$. That pair could NOT be same-side interior angles, option (C).

Answer: (C)

16. HARD · NO CALCULATOR NEEDED

In triangle ABC, the exterior angle at vertex C measures $(7x - 12)°$. This exterior angle equals the sum of the two non-adjacent interior angles, which measure $(3x + 8)°$ and $(2x + 16)°$. What is the value of x?

1. The exterior angle equals the sum of the two remote interior angles: $7x - 12 = (3x + 8) + (2x + 16)$.

2. Combine on the right: $7x - 12 = 5x + 24$.

3. $2x = 36$, so $x = 18$.

4. Check: the exterior angle is $7(18) - 12 = 114°$ and the remote angles are 62° and 52°, which add to 114.

Answer: 18

17. HARD · CALCULATOR

A pole that is 2.4 meters tall casts a shadow 3.2 meters long. At the same time, a nearby communications tower casts a shadow 52 meters long. Using similar triangles, which of the following is closest to the height, in meters, of the tower?

(A) 39.0 ✓

(B) 41.6

(C) 34.7

(D) 36.4

1. The pole with its shadow and the tower with its shadow form similar triangles, so height over shadow is constant. For the pole: $\dfrac{2.4}{3.2} = 0.75$.

2. Tower height $= 0.75 \times 52$ (the tower's shadow).

3. $0.75 \times 52 = 39$ meters. That is option (A).

Answer: (A)

18. HARD · NO CALCULATOR NEEDED

In isosceles triangle ABC, $AB = AC$, and the measure of angle A is $30°$ more than the measure of angle B. What is the measure, in degrees, of angle C?

1. $AB = AC$, so the angles opposite them are equal: angle B = angle C. Call each b.

2. Angle $A = b + 30$, and the three angles add to 180: $(b + 30) + b + b = 180$.

3. $3b = 150$, so $b = 50$ and angle $C = 50°$.

4. Check: $A = 80°$ and $80 + 50 + 50 = 180$.

Answer: 50

19. HARD · NO CALCULATOR NEEDED

In triangle ABC, points M and N are the midpoints of sides AB and AC, respectively. If $MN = 3x - 4$ and $BC = 5x + 2$, what is the length of BC?

(A) 56

(B) 42

(C) 47

(D) 52 ✓

1. The segment joining the midpoints of two sides is parallel to the third side and half as long: $MN = \frac{1}{2}BC$.

2. $3x - 4 = \frac{1}{2}(5x + 2)$. Multiply by 2: $6x - 8 = 5x + 2$.

3. $x = 10$, so $BC = 5(10) + 2 = 52$.

4. Check: $MN = 3(10) - 4 = 26$, which is half of 52. That is option (D).

Answer: (D)

20. HARD · NO CALCULATOR NEEDED

In $\triangle XYZ$, the side lengths are a, b, and c, where $4 < a < 6$ and $9 < b < 11$. Which of the following represents all possible values of c?

(A) $5 < c < 17$

(B) $3 < c < 15$

(C) $3 < c < 17$ ✓

(D) $5 < c < 15$

1. The triangle inequality: $|a - b| < c < a + b$.

2. For the smallest possible lower bound, make $|a - b|$ as small as it can be: a as large as possible and b as small as possible, so $|a - b|$ gets close to $|6 - 9| = 3$ (but never reaches it).

3. For the largest upper bound, make $a + b$ as large as possible: it gets close to $6 + 11 = 17$ (never reaching it).

4. So $3 < c < 17$, option (C). The bounds 5 and 15 come from using other combinations of the end values.

Answer: (C)

21. HARD · NO CALCULATOR NEEDED

In triangle ABC, sides AC and AB are congruent. Points D, C, and B are collinear, with C between D and B. If the measure of angle DCA is $112°$, what is the measure, in degrees, of angle A?

1. D, C, B lie on a line, so angles DCA and ACB are supplementary: $\angle ACB = 180 - 112 = 68°$.

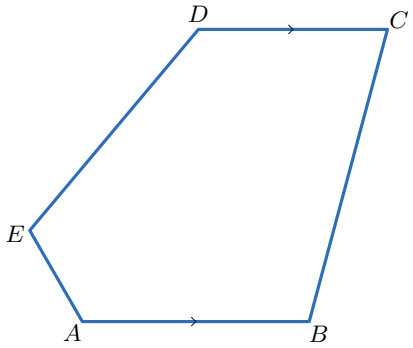
2. $AC = AB$, so the angles opposite them are equal: $\angle B = \angle ACB = 68°$.

3. $\angle A = 180 - 68 - 68 = 44°$.

Answer: 44

22. HARD · NO CALCULATOR NEEDED

In convex pentagon $ABCDE$ shown (not drawn to scale), side $\overline{AB}$ is parallel to side $\overline{DC}$. The measures of four interior angles are $\angle A = (6x)°$, $\angle B = (5x + 5)°$, $\angle C = (3x + 15)°$ and $\angle D = (7x - 10)°$. What is the measure, in degrees, of $\angle E$?



(A) 130

(B) 110 ✓

(C) 70

(D) 105

1. Sides AB and DC are parallel and BC is a transversal, so the same-side interior angles B and C add to $180°$: $(5x + 5) + (3x + 15) = 180$.

2. $8x + 20 = 180$, so $8x = 160$ and $x = 20$.

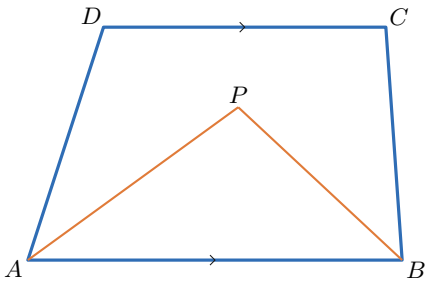
3. Then $\angle A = 120°$, $\angle B = 105°$, $\angle C = 75°$, $\angle D = 130°$.

4. A pentagon's angles add to $(5 - 2) \cdot 180 = 540$, so $\angle E = 540 - (120 + 105 + 75 + 130) = 110°$. Option (B).

Answer: (B)

23. HARD · NO CALCULATOR NEEDED

In trapezoid $ABCD$ shown (not drawn to scale), $\overline{AB} \parallel \overline{DC}$, $\angle D = 108°$, and $\angle C = 94°$. The bisector of $\angle DAB$ and the bisector of $\angle ABC$ intersect at point P inside the trapezoid. What is the measure, in degrees, of the obtuse angle formed at P by these two bisectors?



(A) 79

(B) 22

(C) 108

(D) 101 ✓

1. $AB \parallel DC$, so angles on the same side add to 180°: $\angle A = 180 - 108 = 72°$ and $\angle B = 180 - 94 = 86°$.

2. The bisectors cut these in half: $\angle PAB = 36°$ and $\angle PBA = 43°$.

3. In triangle APB: $\angle APB = 180 - 36 - 43 = 101°$, which is obtuse.

4. The other angle at P is $180 - 101 = 79°$ (acute), so the obtuse angle is 101°, option (D).

Answer: (D)

24. HARD · NO CALCULATOR NEEDED

Two lines intersect at point O, and the smallest of the four angles they form measures 50°. A ray from O bisects each of the four angles. Together with the two lines, this gives eight rays from O, which divide the full turn into eight small angles. A *span* is an angle formed by 1, 2 or 3 consecutive small angles. Which of the following is NOT the measure of any span?

(A) 140° ✓

(B) 115°

(C) 130°

(D) 155°

1. The lines make angles of 50° and $180 - 50 = 130°$. Each bisecting ray halves one of these, giving small angles of $\frac{50}{2} = 25°$ and $\frac{130}{2} = 65°$.

2. Going around O the eight small angles are $25, 25, 65, 65, 25, 25, 65, 65$ degrees (they add to 360).

3. Spans of 1 small angle: 25, 65. Of 2: $25+25 = 50$, $25+65 = 90$, $65+65 = 130$. Of 3: $25+25+65 = 115$, $25+65+65 = 155$.

4. The possible spans are $25, 50, 65, 90, 115, 130, 155$. The values 130, 155 and 115 appear but 140 does not, so the answer is option (A).

Answer: (A)

25. HARD · NO CALCULATOR NEEDED

In equilateral triangle ABC, point D lies on side $\overline{BC}$ such that $BD = \frac{1}{4}BC$. What is the value of $\tan\angle DAC$?

(A) $\frac{3\sqrt{3}}{7}$ **(B)** $\frac{3\sqrt{3}}{5}$ ✓

(C) $\frac{\sqrt{3}}{7}$ (D) $\frac{5\sqrt{3}}{9}$

1. Put the triangle on coordinates: $B = (0,0)$, $C = (4,0)$ and $A = (2, 2\sqrt{3})$ (side 4, so $BD = \frac{1}{4}\cdot 4 = 1$ and $D = (1,0)$).

2. The vectors from A are $\vec{AC} = (2, -2\sqrt{3})$ and $\vec{AD} = (-1, -2\sqrt{3})$.

3. The tangent of the angle between two vectors is $\frac{|\text{cross}|}{\text{dot}}$. Cross: $6\sqrt{3}$. Dot: 10.

4. $\tan\angle DAC = \frac{6\sqrt{3}}{10} = \frac{3\sqrt{3}}{5}$, option (B).

Answer: (B)

26. HARD · NO CALCULATOR NEEDED

In triangle ABC, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that $\overline{DE}$ is parallel to $\overline{BC}$. The measure of $\angle ADE$ is $(4x - 10)°$, the measure of $\angle ABC$ is $(2x + 30)°$, and the measure of $\angle ACB$ is $(x + 25)°$. What is the measure, in degrees, of $\angle BAC$?

(A) 115 (B) 45

(C) 65 ✓ (D) 70

1. $DE \parallel BC$, so the corresponding angles $\angle ADE$ and $\angle ABC$ are equal: $4x - 10 = 2x + 30$.

2. $2x = 40$, so $x = 20$.

3. $\angle ABC = 2(20) + 30 = 70°$ and $\angle ACB = 20 + 25 = 45°$.

4. $\angle BAC = 180 - 70 - 45 = 65°$, option (C).

Answer: (C)

27. HARD · NO CALCULATOR NEEDED

Points A, B, C and D are consecutive vertices of a regular polygon with n sides, and $\overline{AC}$ is a diagonal. The measure of $\angle BAC$ is $(x + 3)°$ and the measure of $\angle ACD$ is $(10x + 15)°$, where x is a constant. What is the value of n?

(A) 12 ✓ (B) 15

(C) 24 (D) 8

1. In a regular n-gon, triangle ABC is isosceles with apex angle equal to the interior angle $180 - \frac{360}{n}$, so $\angle BAC = \frac{180}{n}$ and $\angle ACD = 180 - \frac{540}{n}$ (the interior angle at C minus $\angle BCA$).

2. Set $x + 3 = \frac{180}{n}$ and $10x + 15 = 180 - \frac{540}{n}$. Substitute $x = \frac{180}{n} - 3$: $\frac{1800}{n} - 15 = 180 - \frac{540}{n}$.

3. $\frac{2340}{n} = 195$, so $n = 12$.

4. Check: $\angle BAC = \frac{180}{12} = 15°$ gives $x = 12$, and $10(12) + 15 = 135 = 180 - \frac{540}{12}$. Option (A).

Answer: (A)

28. HARD · NO CALCULATOR NEEDED

A convex polygon has n sides. Its interior angles, listed from smallest to largest, form an arithmetic sequence with smallest angle $120°$ and common difference $5°$. How many sides does the polygon have?

(A) 16 (B) 18

(C) 7 **(D) 9 ✓**

1. The angles are $120, 125, 130, \ldots$, an arithmetic sequence, so their sum is $\frac{n}{2}\big(2\cdot 120+(n-1)\cdot 5\big) = \frac{n}{2}(235+5n)$. An n-gon's angles add to $180(n-2)$.
2. $n(235+5n) = 360(n-2)$ gives $5n^2 - 125n + 720 = 0$, so $n^2 - 25n + 144 = 0$.
3. $(n-9)(n-16) = 0$, so $n = 9$ or $n = 16$.
4. Convex means every angle is under $180°$. The largest angle is $120 + 5(n-1)$: for $n = 16$ it is $195°$ (too big), for $n = 9$ it is $160°$ (fine). So $n = 9$, option (D).

Answer: (D)

29. HARD · NO CALCULATOR NEEDED

Two lines intersect and form four angles. The acute angles each measure $(2k+4)°$, where k is a constant. Each obtuse angle measures 4 degrees more than 3 times the measure of an acute angle. Which of the following CANNOT be the sum, in degrees, of the measures of two of the four angles?

(A) 88 (B) 180

(C) 224 ✓ (D) 272

1. An acute angle and an obtuse angle at an intersection are supplementary: $(2k+4) + \big[3(2k+4)+4\big] = 180$.
2. $4(2k+4)+4 = 180$, so $8k + 20 = 180$ and $k = 20$. The acute angles are $2(20)+4 = 44°$ and the obtuse angles are $3(44)+4 = 136°$.
3. Sums of two of the four angles: acute plus acute $= 88$, acute plus obtuse $= 180$, obtuse plus obtuse $= 272$.
4. The value 224 is not among these, so it CANNOT be a sum: option (C).

Answer: (C)

30. HARD · NO CALCULATOR NEEDED

In triangle ABC, the measure of angle A is $(3x+11)°$, the measure of angle B is $(2x+4)°$, and the measure of angle C is $30°$. What is the value of

$$(\sin A\cos B + \cos A\sin B)(\cos A\cos B - \sin A\sin B)\,?$$

(A) $\frac{1}{4}$ **(B) $-\frac{\sqrt{3}}{4}$ ✓**

(C) $\frac{\sqrt{3}}{4}$ (D) $-\frac{1}{4}$

1. The angles of a triangle add to $180°$, so $A + B = 180 - 30 = 150°$. (The value of x is not needed.)
2. The first factor is $\sin A\cos B + \cos A\sin B = \sin(A+B) = \sin 150° = \frac{1}{2}$.
3. The second factor is $\cos A\cos B - \sin A\sin B = \cos(A+B) = \cos 150° = -\frac{\sqrt{3}}{2}$.
4. Product: $\frac{1}{2}\cdot\left(-\frac{\sqrt{3}}{2}\right) = -\frac{\sqrt{3}}{4}$ (about -0.433), option (B).

Answer: (B)

31. HARD · NO CALCULATOR NEEDED

In triangle ABC, $AB = 12$ and $AC = 18$. The bisector of angle A meets $\overline{BC}$ at D, and $BC = 20$. What is the length of the longer of the two segments BD and DC?

1. The angle bisector theorem: $\dfrac{BD}{DC} = \dfrac{AB}{AC} = \dfrac{12}{18} = \dfrac{2}{3}$.

2. So $BD = 2k$ and $DC = 3k$ with $2k + 3k = BC = 20$, giving $k = 4$.

3. $BD = 8$ and $DC = 12$. The longer segment is 12.

Answer: 12

32. HARD · NO CALCULATOR NEEDED

Two sides of a triangle have lengths 7 and 12. The third side has an integer length. How many different integer lengths are possible for the third side?

1. The third side t must satisfy $12 - 7 < t < 12 + 7$, that is $5 < t < 19$.

2. The integers in that range are $6, 7, \ldots, 18$.

3. Count them: $18 - 6 + 1 = 13$.

Answer: 13

33. HARD · NO CALCULATOR NEEDED

In triangle ABC, $AB = AC$ and angle A measures $40°$. Point D lies on $\overline{AC}$ such that $BD = BC$. What is the measure, in degrees, of angle ABD?

1. Since $AB = AC$ and $\angle A = 40°$, the base angles are $\dfrac{180 - 40}{2} = 70°$ each, so $\angle ABC = \angle C = 70°$.

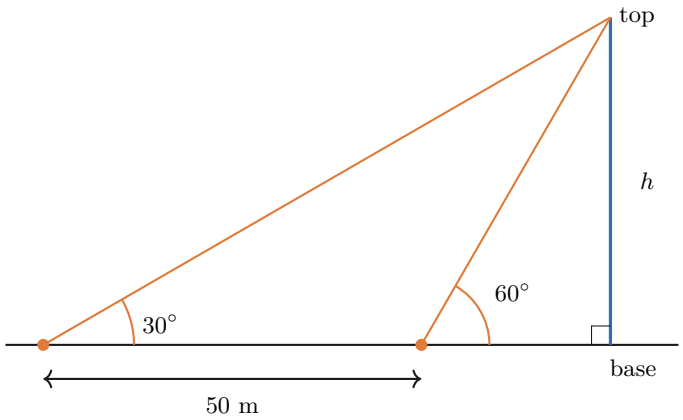
2. In triangle BCD, $BD = BC$ so $\angle BDC = \angle C = 70°$, and $\angle CBD = 180 - 2(70) = 40°$.

3. $\angle ABD = \angle ABC - \angle CBD = 70 - 40 = 30°$.

Answer: 30

34. HARD · CALCULATOR

Engineers surveying a proposed tower measure the angle of elevation to the top of the tower from a point on the ground to be $30°$. Standing 50 meters closer to the base of the tower, along the same straight line, the angle of elevation is $60°$. What is the height of the tower, in meters, to the nearest tenth?



1. Let d be the distance from the nearer point to the base. Then $h = d\tan 60°$ and $h = (d+50)\tan 30°$.

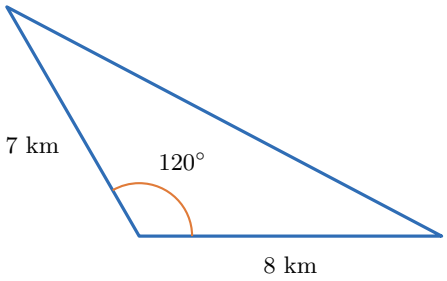
2. Set them equal: $d\sqrt{3} = \dfrac{d+50}{\sqrt{3}}$. Multiply by $\sqrt{3}$: $3d = d + 50$, so $2d = 50$ and $d = 25$.

3. $h = 25\tan 60° = 25\sqrt{3}$. On a calculator, $25 \times 1.7321 = 43.3013$, so $h \approx 43.3$ meters.

Answer: 43.3

35. HARD · NO CALCULATOR NEEDED

A triangular plot of land has two sides of length 7 kilometers and 8 kilometers, with an included angle of $120°$ between them. What is the length, in kilometers, of the third side of the plot?



(A) 15

(B) 11

(C) 12

(D) 13 ✓

1. The third side is opposite the $120°$ angle, so use the Law of Cosines: $c^2 = 7^2 + 8^2 - 2(7)(8)\cos 120°$.

2. $\cos 120° = -\dfrac{1}{2}$, so $c^2 = 49 + 64 - 112\left(-\dfrac{1}{2}\right) = 113 + 56$.

3. $c^2 = 169$, so $c = \sqrt{169} = 13$ kilometers. That is option (D).

Answer: (D)

36. HARD · NO CALCULATOR NEEDED

In right triangle JKL, the right angle is at L. If $\cos(J) = \dfrac{7}{25}$, what is the value of $\sin(J)$?



(A) $\dfrac{24}{25}$ ✓ (B) $\dfrac{25}{7}$

(C) $\dfrac{25}{24}$ (D) $\dfrac{7}{25}$

1. $\cos J = \dfrac{7}{25}$ means the side adjacent to J is 7 and the hypotenuse is 25 (scaled).

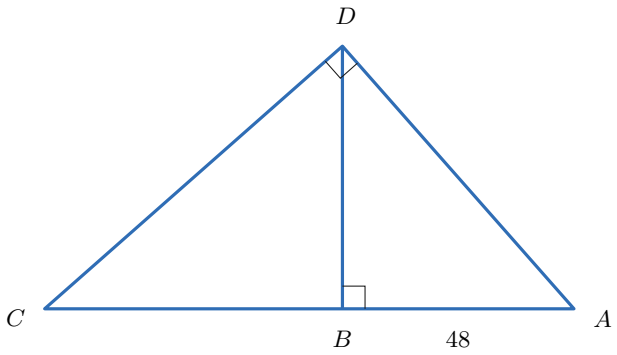
2. The other leg is $\sqrt{25^2 - 7^2} = \sqrt{576} = 24$.

3. $\sin J = \dfrac{\text{opposite}}{\text{hypotenuse}} = \dfrac{24}{25}$. Option (A). (Check by the identity: $\sin^2 J + \cos^2 J = \dfrac{576}{625} + \dfrac{49}{625} = 1$.)

Answer: (A)

37. HARD · NO CALCULATOR NEEDED

In the figure shown, right triangle ACD has a right angle at D, and right triangle BCD, with B on segment AC, has a right angle at B. Both triangles share angle C. If $\cos C = 0.75$ and $AC = 48$, what is the length of BC?



1. Triangle ACD has a right angle at D, so $\cos C = \dfrac{DC}{AC}$ and $DC = 0.75 \times 48 = 36$.

2. Triangle BCD has a right angle at B (B is the foot of the perpendicular from D to AC), and it shares angle C: $\cos C = \dfrac{BC}{DC}$.

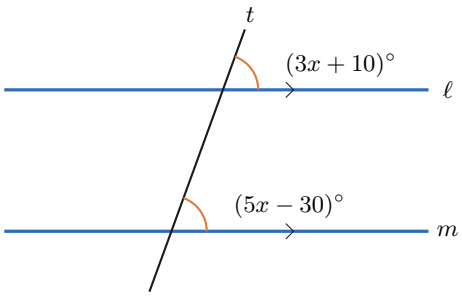
3. $BC = 0.75 \times 36 = 27$.

Answer: 27

PART B · SELF-ASSESSMENT — 24 PROBLEMS

38. Medium · No calculator needed

Line ℓ is parallel to line m and both are intersected by transversal t. One angle formed at the intersection with ℓ measures $(3x+10)°$, and its corresponding angle at the intersection with m measures $(5x-30)°$. What is the value of x?



(A) 15

(B) 20 ✓

(C) 40

(D) 10

1. Corresponding angles formed by a transversal across parallel lines are equal, so $3x + 10 = 5x - 30$.

2. Subtract $3x$ and add 30 to both sides: $10 + 30 = 5x - 3x$, so $40 = 2x$.

3. Divide by 2: $x = 20$.

4. Check: $3(20) + 10 = 70$ and $5(20) - 30 = 70$, equal as required. That is option (B).

Answer: (B)

39. Medium · No calculator needed

In triangle ABC, the measure of angle A is twice the measure of angle B, and the measure of angle C is 24° more than the measure of angle B. What is the measure of angle A, in degrees?

(A) 39

(B) 52

(C) 78 ✓

(D) 87

1. Let angle $B = x$. Then angle $A = 2x$ and angle $C = x + 24$.

2. The angles of a triangle add to 180°: $2x + x + (x + 24) = 180$, so $4x + 24 = 180$.

3. $4x = 156$, so $x = 39$ and angle $A = 2(39) = 78°$.

4. Check: $78 + 39 + 63 = 180$. That is option (C); the value 39 is angle B, the other options come from answering for the wrong angle or an arithmetic slip.

Answer: (C)

40. Medium · No calculator needed

A regular polygon has exterior angles that each measure 24°. How many sides does the polygon have?

(A) 15 ✓ (B) 18

(C) 24 (D) 12

1. The exterior angles of any convex polygon add up to 360°.

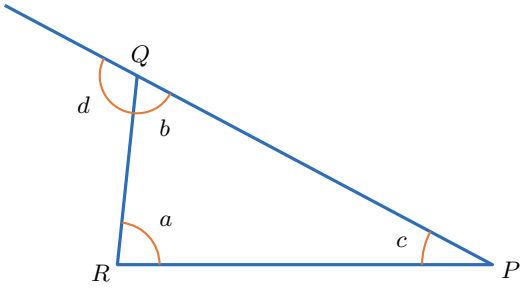
2. In a regular polygon they are all equal, so the number of sides is $\frac{360}{24}$.

3. $\frac{360}{24} = 15$ sides. That is option (A).

Answer: (A)

41. Medium · No calculator needed

In the figure shown, a, b and c are the three interior angles of a triangle, with $a = 3c$. Side PQ of the triangle is extended through Q, and the angle d that this extension makes with the side at Q measures 112°, so b and d together form a straight line. What is the value of $a + b$?



(A) 158 (B) 140

(C) 146 **(D)** 152 ✓

1. Angles b and d form a straight line, so $b = 180° - 112° = 68°$.

2. The interior angles add to 180°: $a + b + c = 180$ with $a = 3c$ and $b = 68$, so $3c + 68 + c = 180$ and $4c = 112$.

3. $c = 28°$ and $a = 3(28) = 84°$.

4. $a + b = 84 + 68 = 152$. That is option (D).

Answer: (D)

42. Medium · No calculator needed

What is the positive difference, in degrees, between the measure of an exterior angle of a regular octagon (8 sides) and the measure of an exterior angle of a regular decagon (10 sides)?

1. Every exterior angle of a regular n-gon is $\frac{360°}{n}$ (the exterior angles of a polygon sum to 360°).

2. Octagon: $\frac{360}{8} = 45°$. Decagon: $\frac{360}{10} = 36°$.

3. The positive difference is $45 - 36 = 9$.

Answer: 9

43. Medium · No calculator needed

At a certain time of day, a 6-meter flagpole casts a shadow that is 4 meters long. At the same time, a nearby building casts a shadow that is 30 meters long. What is the height, in meters, of the building?

1. The flagpole with its shadow and the building with its shadow form similar right triangles, so height over shadow is the same: $\frac{6}{4} = \frac{h}{30}$.
2. Multiply both sides by 30: $h = \frac{6}{4} \times 30 = 1.5 \times 30$.
3. $h = 45$ meters. Check: $\frac{45}{30} = 1.5 = \frac{6}{4}$.

Answer: 45

44. Medium · No calculator needed

Triangle DEF is similar to triangle GHI, where D, E, and F correspond to G, H, and I, respectively. The measure of angle D is $48°$, and the measure of angle E is $63°$. What is the measure of angle H?

(A) $117°$ (B) $48°$

(C) $63°$ ✓ (D) $69°$

1. Similar triangles have equal corresponding angles, whatever the scale factor.
2. $D \leftrightarrow G$, $E \leftrightarrow H$ and $F \leftrightarrow I$, so angle H equals angle $E = 63°$.
3. Check: angle $F = 180 - 48 - 63 = 69°$, so triangle GHI has angles $48°$, $63°$ and $69°$. That is option (C). (The value $69°$ is angle F and $117°$ is $180 - 63$.)

Answer: (C)

45. Medium · No calculator needed

In a triangle, the measures of the three interior angles are x, $2x$, and $(3x - 30)$ degrees. What is the value of x?

(A) 32 **(B) 35 ✓**

(C) 40 (D) 30

1. The three angles of a triangle add to $180°$: $x + 2x + (3x - 30) = 180$.
2. Combine like terms: $6x - 30 = 180$, so $6x = 210$.
3. $x = 35$. Check: the angles are $35°$, $70°$ and $75°$, which add to 180. That is option (B).

Answer: (B)

46. Medium · No calculator needed

Line ℓ is parallel to line m, and a transversal crosses both lines. One angle formed at line ℓ measures $(4x+5)°$, and the corresponding angle formed at line m measures $(6x-21)°$. What is the value of x?

(A) 15 (B) 18

(C) 11 **(D) 13 ✓**

1. Corresponding angles across parallel lines are equal: $4x + 5 = 6x - 21$.

2. Move the x terms to one side and the numbers to the other: $5 + 21 = 6x - 4x$, so $26 = 2x$.

3. $x = 13$. Check: $4(13) + 5 = 57$ and $6(13) - 21 = 57$. That is option (D).

Answer: (D)

47. Medium · No calculator needed

Two lines intersect, forming four angles. One angle measures $(5x + 10)°$ and the angle vertical to it measures $(3x + 30)°$. What is the value of x?

(A) 10 ✓ (B) 12

(C) 15 (D) 8

1. Vertical angles are equal: $5x + 10 = 3x + 30$.

2. Subtract $3x$ and 10: $2x = 20$.

3. $x = 10$. Check: $5(10) + 10 = 60$ and $3(10) + 30 = 60$. That is option (A).

Answer: (A)

48. Medium · No calculator needed

A triangle has side lengths of 9, 14, and x, where x is an integer. What is the greatest possible value of x?

(A) 21 **(B) 22 ✓**

(C) 23 (D) 24

1. The triangle inequality says the third side must be less than the sum of the other two: $x < 9 + 14 = 23$.

2. x is an integer and $x < 23$ (it cannot equal 23), so the greatest possible value is 22.

3. Check the other two inequalities: $9 + 22 = 31 > 14$ and $14 + 9 = 23 > 22$. Option (B). (The value 23 is the sum, which would make a flat triangle.)

Answer: (B)

49. Medium · No calculator needed

Triangles ABC and DEF are similar. Side AB of triangle ABC corresponds to side DE of triangle DEF. If the length of $\overline{AB}$ is 12 units and the length of $\overline{DE}$ is 8 units, then the area of triangle DEF is equal to

(A) $\frac{16}{9}$ (area of triangle ABC)

(B) $\frac{2}{3}$ (area of triangle ABC)

(C) $\frac{4}{9}$ (area of triangle ABC) ✓

(D) $\frac{9}{4}$ (area of triangle ABC)

1. The scale factor from ABC to DEF is the ratio of corresponding sides: $\frac{DE}{AB} = \frac{8}{12} = \frac{2}{3}$.

2. Areas of similar figures scale by the square of the scale factor: $\left(\frac{2}{3}\right)^2 = \frac{4}{9}$.

3. So area of $DEF = \frac{4}{9}$ (area of ABC), option (C). Since DEF is the smaller triangle the ratio must be less than 1, which rules out the two fractions above 1.

Answer: (C)

50. Medium · No calculator needed

Two parallel lines are cut by a transversal. A pair of alternate interior angles measure $(6x - 8)°$ and $(4x + 12)°$. What is the measure, in degrees, of each of these angles?

1. Alternate interior angles between parallel lines are equal: $6x - 8 = 4x + 12$.

2. $2x = 20$, so $x = 10$.

3. Each angle is $6(10) - 8 = 52°$. Check: $4(10) + 12 = 52$.

Answer: 52

51. Medium · No calculator needed

An exterior angle of a triangle measures $(8x-6)°$. The two remote interior angles measure $(3x+14)°$ and $(2x+19)°$. What is the measure, in degrees, of the exterior angle?

1. The exterior angle equals the sum of the remote interior angles: $8x - 6 = (3x + 14) + (2x + 19) = 5x + 33$.

2. $3x = 39$, so $x = 13$.

3. The exterior angle is $8(13) - 6 = 98°$. Check: $53 + 45 = 98$.

Answer: 98

52. Medium · No calculator needed

In an isosceles triangle, the two base angles each measure $(x+20)°$ and the vertex angle measures $(4x-10)°$. What is the measure, in degrees, of the vertex angle?

1. The three angles add to 180: $(4x - 10) + 2(x + 20) = 180$.

2. $6x + 30 = 180$, so $6x = 150$ and $x = 25$.

3. The vertex angle is $4(25) - 10 = 90°$. Check: the base angles are $45°$ each and $90 + 45 + 45 = 180$.

Answer: 90

53. Medium · No calculator needed

In triangle ABC, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ such that $\overline{DE}$ is parallel to $\overline{BC}$. If $AD = 10$, $DB = 6$ and $AE = 15$, what is the length of $\overline{EC}$?

1. $DE \parallel BC$, so the sides are proportional: $\dfrac{AD}{AB} = \dfrac{AE}{AC}$. Here $AB = AD + DB = 10 + 6 = 16$.

2. $\dfrac{10}{16} = \dfrac{15}{AC}$, so $AC = \dfrac{15 \cdot 16}{10} = 24$.

3. $EC = AC - AE = 24 - 15 = 9$.

Answer: 9

54. Medium · No calculator needed

Each interior angle of a regular polygon measures $140°$. How many sides does the polygon have?

1. Each exterior angle is $180 - 140 = 40°$ (interior plus exterior make a straight angle).

2. The exterior angles of a polygon add to $360°$, and in a regular polygon they are all equal, so the number of sides is $\dfrac{360}{40}$.

3. $\dfrac{360}{40} = 9$ sides. Check: $\dfrac{180(9-2)}{9} = 140°$.

Answer: 9

55. Medium · No calculator needed

A trapezoid has parallel sides of length 10 and 24. A segment joins the midpoints of the two non-parallel sides. What is the length of this segment?

(A) 17 ✓ (B) 34

(C) 14 (D) 7

1. The segment joining the midpoints of the non-parallel sides (the midsegment) is parallel to the bases and has length equal to their average.

2. Add the bases: $10 + 24 = 34$.

3. Divide by 2: $\dfrac{34}{2} = 17$. Option (A). (The value 34 is the sum of the bases, not their average.)

Answer: (A)

56. Medium · No calculator needed

In the xy-plane, a triangle has vertices $(0, 0)$, $(8, 0)$ and $(3, k)$, where $k > 0$. The area of the triangle is 20. What is the value of k?

1. The base on the x-axis is from $(0, 0)$ to $(8, 0)$, so the base is 8. The height is the y-coordinate k of the third vertex (the x-coordinate 3 does not matter).

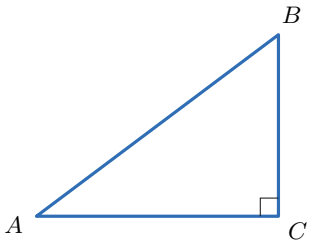
2. Area $= \dfrac{1}{2}(8)(k) = 4k$, so $4k = 20$.

3. $k = 5$.

Answer: 5

57. Medium · No calculator needed

In right triangle ABC, the right angle is at C. If $\sin(A) = \frac{3}{5}$, what is the value of $\cos(B)$?



(A) $\frac{4}{5}$

(B) $\frac{5}{3}$

(C) $\frac{3}{4}$

(D) $\frac{3}{5}$ ✓

1. In a right triangle the two acute angles are complementary: $A + B = 90°$.

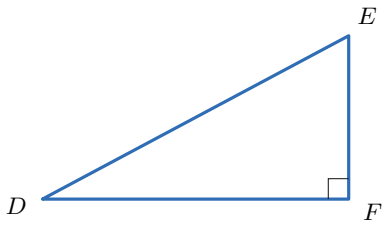
2. Use a 3-4-5 triangle with $\sin A = \frac{3}{5}$: the side opposite A is 3, the hypotenuse is 5, and the other leg is 4.

3. $\cos B = \frac{\text{side adjacent to } B}{\text{hypotenuse}}$. The side adjacent to B is the same side that is opposite A, namely 3, so $\cos B = \frac{3}{5}$. That is option (D): $\sin A = \cos(90° - A)$.

Answer: (D)

58. Medium · No calculator needed

In right triangle DEF, the right angle is at F. If $\tan(D) = \frac{8}{15}$, what is the value of $\tan(E)$?



(A) $\frac{17}{15}$

(B) $\frac{8}{15}$

(C) $\frac{15}{8}$ ✓

(D) $\frac{8}{17}$

1. In a right triangle DEF with the right angle at F, angles D and E are complementary, and the two legs swap roles: the leg opposite D is adjacent to E.

2. $\tan D = \frac{\text{opposite } D}{\text{adjacent } D} = \frac{8}{15}$, so $EF = 8$ and $DF = 15$ (and $DE = \sqrt{8^2 + 15^2} = 17$).

3. For angle E: the opposite side is $DF = 15$ and the adjacent side is $EF = 8$, so $\tan E = \frac{15}{8}$ (the reciprocal of $\tan D$). Option (C).

Answer: (C)

59. Medium · No calculator needed

If θ is an acute angle and $\sin(\theta) = \dfrac{5}{13}$, what is the value of $\tan(\theta)$?

(A) $\dfrac{12}{13}$

(B) $\dfrac{5}{12}$ ✓

(C) $\dfrac{12}{5}$

(D) $\dfrac{5}{13}$

1. $\sin\theta = \dfrac{5}{13}$ means the opposite side is 5 and the hypotenuse is 13.

2. The adjacent side is $\sqrt{13^2 - 5^2} = \sqrt{144} = 12$ (a 5-12-13 triangle).

3. $\tan\theta = \dfrac{\text{opposite}}{\text{adjacent}} = \dfrac{5}{12}$. Option (B).

Answer: (B)

60. Medium · No calculator needed

In a 30°-60°-90° right triangle, the side opposite the 30° angle has length 7. What is the length of the hypotenuse?

(A) 7

(B) $7\sqrt{2}$

(C) $7\sqrt{3}$

(D) 14 ✓

1. In a 30°-60°-90° triangle the sides are in the ratio $x : x\sqrt{3} : 2x$, where x is opposite the 30° angle.

2. Here $x = 7$.

3. The hypotenuse is $2x = 2(7) = 14$. Option (D).

Answer: (D)

61. Medium · No calculator needed

What is the value of $\cos(60°) + \sin(30°)$?

(A) 1 ✓

(B) $\sqrt{3}$

(C) $\dfrac{1}{2}$

(D) $\dfrac{\sqrt{3}}{2}$

1. Use the special-angle values: $\cos 60° = \dfrac{1}{2}$ and $\sin 30° = \dfrac{1}{2}$ (these are equal because $60 + 30 = 90$).

2. Add: $\dfrac{1}{2} + \dfrac{1}{2} = 1$.

3. The sum is 1, option (A).

Answer: (A)